

This assignment is about a program that is capable of playing the Nim game described in class. Implement the **alpha beta search** algorithm in C++ or in Java on this game. Assume the cutoff test is always false. The input of the resulting program should be the current position represented in the following format:

$n \ p_1 \ p_2 \ \dots \ p_n$

Where n is the number of packs in the current position and $p_1 \ p_2 \ \dots \ p_n$ are the number of sticks in these packs.

The output of the algorithm should contain:

- List of successors of the current state with their evaluations
- The best successor.
- The number of pruning events.
- The number of states evaluated.

For example, the program should work as follows:

Enter current position: 1 5

Output:

Successors:

1 4, value=+1

2 3, value=-1

The best successor is 1 4

The number of prunings = 0

The number of states evaluated = 6

The best successor state from this state is 1 4

Your program should be well documented and must contain separate functions for terminal test, utility, min, max, expansion.